CONVOLUTIONAL NEURAL NETWORKS BUILDING AN IMAGE CLASSIFIER WITH ARCHITECTURAL OPTIMIZATION AND REGULARIZATION

1. Introduction

Image classification is the most basic yet widely researched problem of computer vision. Image classification entails attaching a tag to an image based on visual data. Conventional machine learning approaches face issues with image data because of its huge dimensionality and complex spatial arrangements. The introduction of deep learning has helped to considerably enhance performance levels, especially CNNs (Convolutional Neural Networks).

CNNs are uniquely built to analyze grid-like data as images. These systems begin learning the hierarchical representations of features that range from low-level edges and depths to high-level object features. CNNs are powerful to the extent that they can classify images.

This tutorial covers the influence of CNN architecture design and regularization techniques on performance. To gain insights into the impact of increasing model complexity and applying regularisation, we compare three CNN models instead of building a single one.

The experiments have been done using the CIFAR-10 dataset which is one of the benchmark datasets. It has a total of 60,000 colour images involving 10 classes which include airplane, automobile, animal, and ship classes.

2. Understanding Convolutional Neural Networks

CNNs are composed of several key components that enable them to extract meaningful features from images:

Convolutional Layers: These layers apply filters (kernels) across the input image to detect patterns such as edges, textures, and shapes. Each filter produces a feature map that highlights specific visual characteristics.

Activation Functions: Non-linear activation functions such as ReLU (Rectified Linear Unit) introduce non-linearity into the network, allowing it to learn complex relationships.

Pooling Layers: Pooling reduces the spatial dimensions of feature maps, helping to decrease computational cost and improve generalization. Max pooling is commonly used to retain the most important features.

Fully Connected Layers: These layers perform classification based on extracted features. They map high-level representations to output class probabilities.

3. Challenges in Training CNNs

Notwithstanding the efficacy, CNNs do face challenges.

- **Overfitting:** Deep networks learn by heart rather than generalizing.
- **Vanishing gradients**: Deep layers won't learn because the gradients will be again vanishing.
- **Training instability:** The performance of the model can vary.

Architectural improvements and regularization techniques will help to overcome them.

4. Improving CNN Performance

This tutorial focuses on two main strategies:

Increasing Model Depth: Deeper networks can capture more complex patterns and improve feature extraction. However, they also increase the risk of overfitting.

Regularization Techniques:

- Dropout: Randomly disables neurons during training to prevent reliance on specific features.
- Batch Normalization: Stabilizes learning by normalizing layer inputs.

5. Experimental Setup

Dataset:

The CIFAR-10 dataset contains:

- 60,000 images (32×32 resolution)
- 10 classes
- Balanced distribution

The data was normalized to a range of [0,1], and labels were converted into categorical format.

Model Architectures:

Model 1: Baseline CNN

- Simple architecture
- Two convolutional layers
- No regularization

Model 2: Improved CNN

- Deeper architecture
- Additional convolutional layers
- Enhanced feature extraction capability

Model 3: Regularized CNN

- Includes Batch Normalization
- Includes Dropout
- Designed to reduce overfitting and improve generalization

6. Results and Analysis

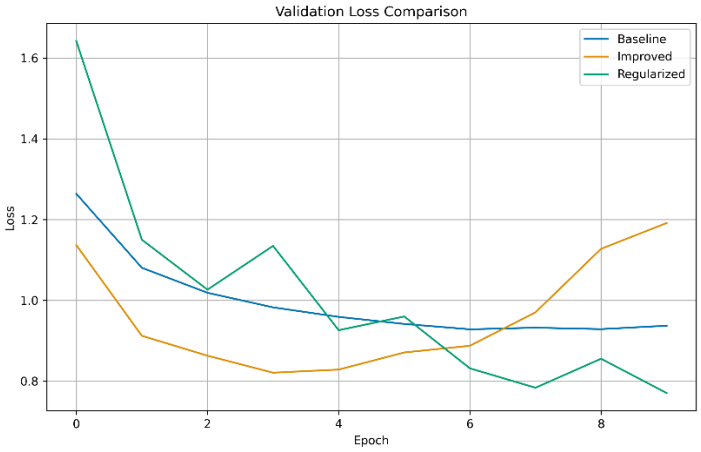
Sample Dataset Visualization: The first figure shows example images from the dataset. These images highlight the diversity and complexity of classes, ranging from animals to vehicles. Some classes share similar visual features, making classification challenging.

Validation Loss Comparison

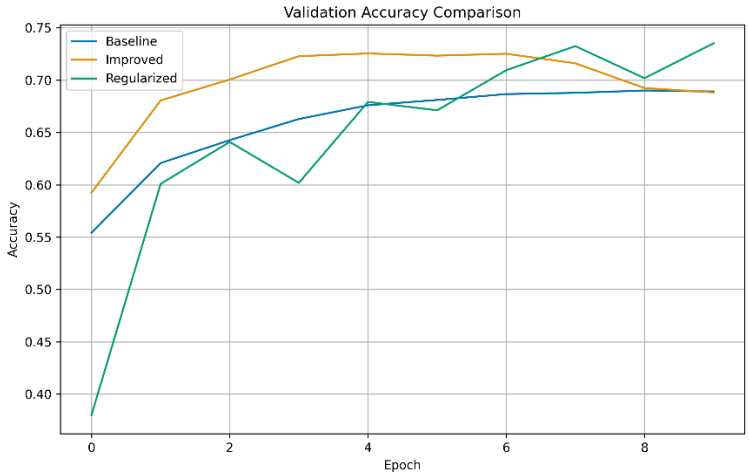
The plot of validation loss compares how well each model reduces error over time.

- The baseline model converges slower, but loss values are greater.
- The enhanced model's depth allows for more effective loss reduction.
- The regularized model shows lowest and most stable loss.

Due to regularization, model performs better and avoids overfitting with training data.



Validation Accuracy Comparison:



The accuracy plot reveals key performance differences:

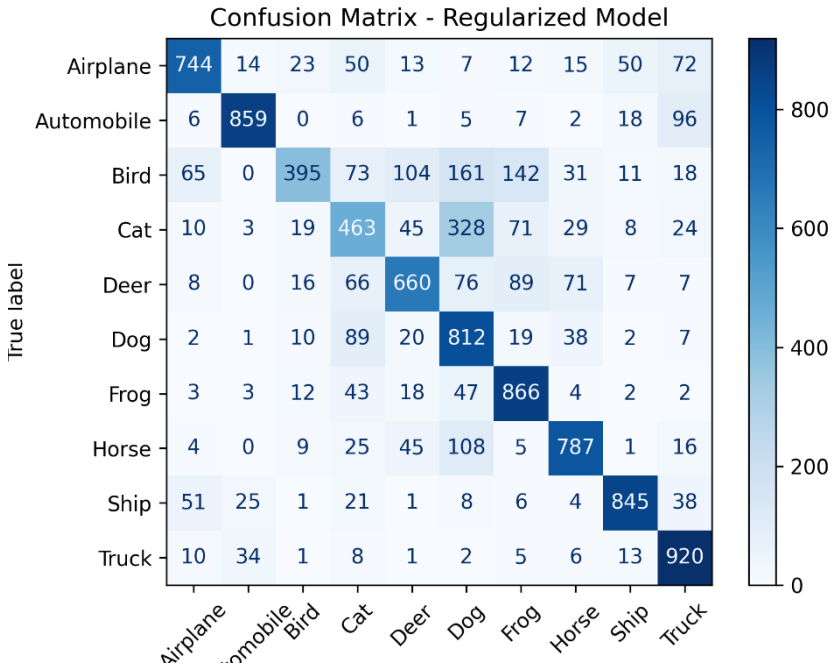
- The baseline model achieves moderate accuracy but plateaus early.
- The improved model achieves higher accuracy due to better feature learning.
- The regularized model achieves the highest and most stable accuracy.

The smoother curve of the regularized model suggests improved training stability.

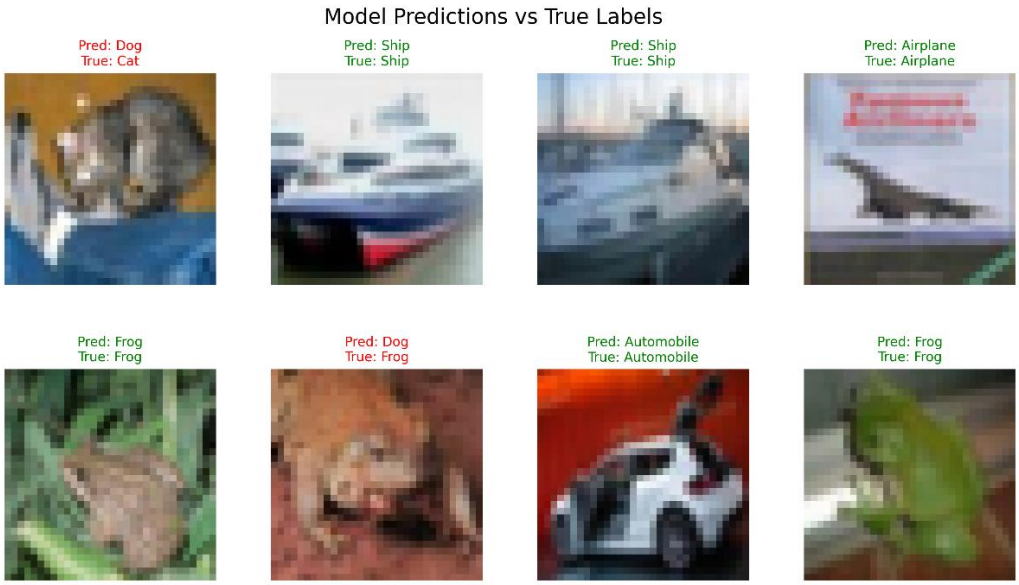
Confusion Matrix Analysis:
The confusion matrix provides class-wise performance evaluation.

- High accuracy is observed in classes such as **automobiles and ships**, which have distinct shapes.
- Misclassifications occur between visually similar classes such as:
    - Cats and dogs
    - Frogs and deer

This indicates that the model struggles with fine-grained distinctions between similar objects.



Prediction Visualization:



The final figure presents a grid of predictions compared with true labels.

- Correct predictions are highlighted in green.
- Incorrect predictions are highlighted in red.

The model performs well on well-organized objects, but poorly on ambiguous or low-resolution images. Such as.

- A cat misclassified as a dog
- A frog misclassified as a dog

This shows that while the model captures general patterns, it still struggles to differentiate small visual differences.

7. Discussion:

The performance of CNN discussed in detail is greatly influenced by the architecture and the regularized used. The baseline model is a good reference point but is not capable of learning complex patterns. It doesn't extract too many features giving it a smooth curve.

The new model fixes the above disadvantage by being deeper and learning richer representations. As a result, there is a noticeable improvement in accuracy and loss. Nevertheless, having a deeper architecture increases the risk of overfitting, that is, it may start memorizing the training data.

The regularized model performs best on average. Batch normalization and dropout both help in training a neural network, batch normalization reduces the internal covariance shift and dropout prevents overfitting by dropping hidden neurons. Consequently, this leads to smoother learning curves and better generalization.

Another important observation is the complexity-performance trade-off. Deeper and regularized models are efficient but costly in terms of computation. Striking the right balance between model complexity and efficiency is key.

It is useful from the application point of view, says paper. Models must be both robust and generalizable in areas like medical imaging and self-driving cars. Regularization techniques help us in preventing models from becoming too complex and overfitting.

Yet limitations persist. The classification task is difficult due to low-resolution images in the CIFAR-10 dataset. The models utilized in this tutorial are also relatively simple in comparison to the state-of-the-art architectures, e.g. ResNet or EfficientNet.

8. Conclusion

- CNNs are highly effective for image classification tasks.
- Increasing model depth improves feature extraction and accuracy.
- Regularization techniques such as Batch Normalization and Dropout significantly enhance generalization.
- The regularized model achieved the best performance with stable learning behaviour.
- Misclassifications highlight the challenge of distinguishing visually similar classes.
- The findings show that architectural enhancement and regularization result in reliable and scalable deep learning methods.
- The findings of this study promote the need to establish models with the dual goals of attaining accuracy and stability.
- All in all, this experiment shows model design is important to achieve the best performance on practical image classification tasks.

9. Future Work

Use advanced architectures like ResNet or EfficientNet.
Utilize data augmentation methods (rotation, flipping).
Using finer resolution dataset will help.
Make use of transfer learning via pre-trained models.
Enhance the understandability of your model results.

10. References:

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- CIFAR-10 Dataset: https://www.cs.toronto.edu/~kriz/cifar.html